

Monte Carlo Test Statistic Distributions:

Kolmogorov-Smirnov, Durbin-Watson, Anderson-Darling

Terrence P. Lenaghan

Project Group 293

Abstract

Statistical hypothesis testing is dependent on critical values establishing significance thresholds. The Kolmogorov-Smirnov, Durbin-Watson, and Anderson-Darling test statistics lack closed-form analytical solutions, which forces researchers to rely on incomplete lookup tables. This project employs Monte Carlo simulation with 10^7 iterations per configuration to generate complete probability distributions and critical values at the 0.75, 0.90, 0.95, and 0.99 quantile levels. The application includes five sample sizes ($n = 30, 50, 100, 500, 1000$) for each test statistic resulting in 15 configurations. Validation targets include mean absolute error below 0.002 for quantile estimates and Type I error rates within ± 0.005 of significance levels. The computational framework achieves quantile estimates with four-decimal precision through convergence monitoring (0.0001 tolerance threshold). This application could enable researchers to obtain exact critical values for any sample size configuration in clinical trials, quality control, or econometric modeling applications.

Background and Description of the Problem

Introduction and Motivation

Hypothesis testing relies on critical values that are derived from test statistic distributions. These are used to establish significance thresholds. When a test statistic exceeds its critical value at a chosen α level, researchers reject the null hypothesis. This decision process requires precise knowledge of the distribution that many complex test statistics lack in closed form (Cheng et al., 2023).

Three widely applied test statistics present this challenge: the Kolmogorov-Smirnov (K-S) statistic for goodness-of-fit testing, the Durbin-Watson (D-W) statistic for autocorrelation detection, and the Anderson-Darling (A-D) statistic for tail-sensitive distributional assessment. Published tables for these statistics cover limited sample sizes and significance levels, forcing researchers to use either table values or imprecise p -values. Clinical trials, quality control applications, and econometric modeling all depend on accurate critical values that these current resources cannot provide.

Monte Carlo simulation offers a solution. By generating millions of random samples under null conditions, empirical distributions emerge with arbitrary precision at any quantile (Cowan, 2024). This computational approach handles problems too complex for exact mathematical solutions and reveals small-sample behavior that approximations miss.

Literature Review

Monte Carlo Simulation Methods

The tidy simulation framework developed by van Kesteren (2025) organizes Monte Carlo studies into four components: simulation grid, generation function, analysis function, and results table. The simulation grid specifies all factor combinations of sample sizes and iteration counts. The generation function samples data from defined processes. The analysis function computes the statistics, and the results table aggregates outputs for analysis.

Standardized metrics are used in performance assessment for simulation studies. Bias can measure the difference between estimated and true parameter values. Standard error quantifies variability across simulation iterations. Mean squared error (MSE) combines bias and variance into a single accuracy measure: $\text{MSE} = (\hat{\beta} - \beta)^2 + \text{SE}(\hat{\beta})^2$ (Banack et al., 2021). Coverage probability assesses if confidence intervals contain true values at nominal rates.

Convergence diagnostics determine when a simulation has achieved precision. The Gelman-Rubin $\hat{R}$ statistic compares within-chain and between-chain variance across independent runs; values below 1.1 indicate convergence (van Kesteren, 2025). Reproducibility requires deterministic pseudo-random number generation in algorithms such as the Mersenne Twister, which provides a period of $2^{19937} - 1$ (Cowan, 2024; Antunes and Hill, 2024).

Test Statistics Under Investigation

Kolmogorov-Smirnov Statistic. The K-S statistic measures maximum deviation between empirical and theoretical cumulative distribution functions:

$$D_n = \sup_x |F_n(x) - F(x)| \quad (1)$$

where $F_n(x)$ denotes the empirical CDF and $F(x)$ the theoretical CDF. The asymptotic critical value follows $c(\alpha) = \sqrt{-\frac{1}{2} \ln(\alpha/2)}$, though finite-sample corrections improve accuracy for small n . Applications include distribution validation and risk modeling.

Durbin-Watson Statistic. The D-W statistic detects first-order autocorrelation in regression residuals:

$$DW = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2} \quad (2)$$

Values range from 0 to 4, with 2.0 indicating no autocorrelation. Time series analysis and econometric modeling require precise critical bounds that published tables provide only for specific configurations.

Anderson-Darling Statistic. The A-D statistic emphasizes tail deviations through weighted integration:

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^n (2i-1) [\ln(\Phi(z_i)) + \ln(1 - \Phi(z_{n+1-i}))] \quad (3)$$

Finite-sample adjustment applies: $A_{\text{adj}}^2 = A^2(1 + 4/n - 25/n^2)$. Finance and reliability engineering employ this test for extreme value analysis where tail behavior determines risk.

Research Objectives

This project has four specific aims: (1) generate complete empirical distributions for K-S, D-W, and A-D statistics across sample sizes $n \in \{30, 50, 100, 500, 1000\}$; (2) estimate quantiles at 0.75, 0.90, 0.95, and 0.99 levels with mean absolute error below 0.001; (3) validate Type I error rates within ± 0.005 of nominal levels; and (4) ensure reproducibility to four decimal places through controlled random number generation.

Report Organization

Section 2 describes the simulation methodology, implementation details, and validation procedures. Section 3 presents empirical distributions, quantile tables, and validation outcomes. Section 4 interprets findings and acknowledges limitations. Section 5 summarizes conclusions and identifies future work that may be needed.

Main Findings

Problem Statement and Prior Work

Statistical hypothesis testing requires critical values that determine rejection thresholds. For complex test statistics like Kolmogorov-Smirnov (K-S), Durbin-Watson (D-W), and Anderson-Darling (A-D), closed-form distributions do not exist, forcing researchers to depend on incomplete lookup tables from decades-old publications (Banack et al., 2021). These tables cover only specific sample sizes and significance levels. When needed values fall outside published ranges, analysts manually insert values or accept imprecise p -values (?). Many researchers select less appropriate tests because distributions for their preferred test remain undocumented.

Monte Carlo simulation addresses this limitation by generating empirical distributions through sampling (Cowan, 2024). The approach produces any quantile for any sample size, replacing static tables with dynamic functions. Today’s computing power can complete millions of iterations in hours versus the weeks required when the original tables were constructed.

Implementation

The simulation framework executes at least 10^7 iterations per configuration across three test statistics and five sample sizes ($n \in \{30, 50, 100, 500, 1000\}$), and outputs 15 total configurations. Target quantiles include 0.75, 0.90, 0.95, and 0.99. All simulations sample from the standard normal distribution $N(0, 1)$ with random seed 42 for reproducibility.

Each test statistic follows its standard formulation. K-S computes $D = \max(D^+, D^-)$ where $D^+ = \max_i(\hat{F}_n(x_i) - F(x_i))$ and $D^- = \max_i(F(x_i) - \hat{F}_{n-1}(x_i))$. D-W calculates $DW = \sum(\Delta e_t)^2 / \sum e_t^2$ with first differences $\Delta e_t = e_t - e_{t-1}$. A-D applies weighted sums on sorted, standardized observations with numerical safeguards to prevent $\log(0)$ errors.

The computational architecture employs batch processing with 100,000-iteration checkpoints. Convergence requires quantile changes below 0.0001 between consecutive batches, with minimum 10 batches before assessment.

Program Execution

The Python implementation requires NumPy, SciPy, h5py, and Matplotlib. Configuration parameters load from YAML files specifying test statistics, sample sizes, and iteration counts. The entry point `scripts/run_production_simulations.py` launches all simulations. Checkpoint recovery enables resumption of interrupted runs. Outputs write to `data/processed/` with figures exported to `reports/figures/`.

Results

All 15 configurations converged with $\hat{R} < 1.05$ across quantiles. Total computation required approximately 54 minutes at 47,609 iterations per second. Table 1 presents critical value estimates with four-decimal precision.

Table 1: Critical values for all test statistics

statistic	sample size	q75	q90	q95	q99
Kolmogorov Smirnov	30	0.1805	0.2172	0.2412	0.2892
Kolmogorov Smirnov	50	0.1409	0.1694	0.1877	0.2243
Kolmogorov Smirnov	100	0.1002	0.1206	0.1340	0.1604
Kolmogorov Smirnov	500	0.0452	0.0543	0.0604	0.0726
Kolmogorov Smirnov	1000	0.0321	0.0386	0.0427	0.0513
Durbin Watson	30	2.1763	2.3900	2.5167	2.7496
Durbin Watson	50	2.1505	2.3208	2.4187	2.6007
Durbin Watson	100	2.1159	2.2367	2.3086	2.4404
Durbin Watson	500	2.0567	2.1109	2.1433	2.2034
Durbin Watson	1000	2.0410	2.0796	2.1024	2.1461
Anderson Darling	30	0.4591	0.6174	0.7359	1.0091
Anderson Darling	50	0.4630	0.6212	0.7376	1.0130
Anderson Darling	100	0.4662	0.6266	0.7435	1.0206
Anderson Darling	500	0.4702	0.6286	0.7462	1.0323
Anderson Darling	1000	0.4702	0.6318	0.7503	1.0285

Validation confirmed statistical accuracy through three mechanisms. Theoretical comparison yielded mean absolute error below 0.002 for all statistics at $n \geq 50$. Type I error validation generated 10,000 independent null samples, with empirical rejection rates falling within ± 0.005 of nominal levels (0.25, 0.10, 0.05, 0.01). Cross-validation split simulations into five independent chains of 2 million iterations each. The Gelman-Rubin diagnostic returned $\hat{R} < 1.1$ for all configurations, with coefficient of variation below 0.01 across chains.

Distribution characteristics matched theoretical expectations. K-S statistics exhibited right skew bounded in $[0, 1]$ with variance decreasing as n increased. D-W statistics centered near 2.0 with approximate symmetry and bounds $[0, 4]$. A-D statistics displayed heavy right tails unbounded above. Figure 1 displays histograms with kernel density estimates for each configuration.

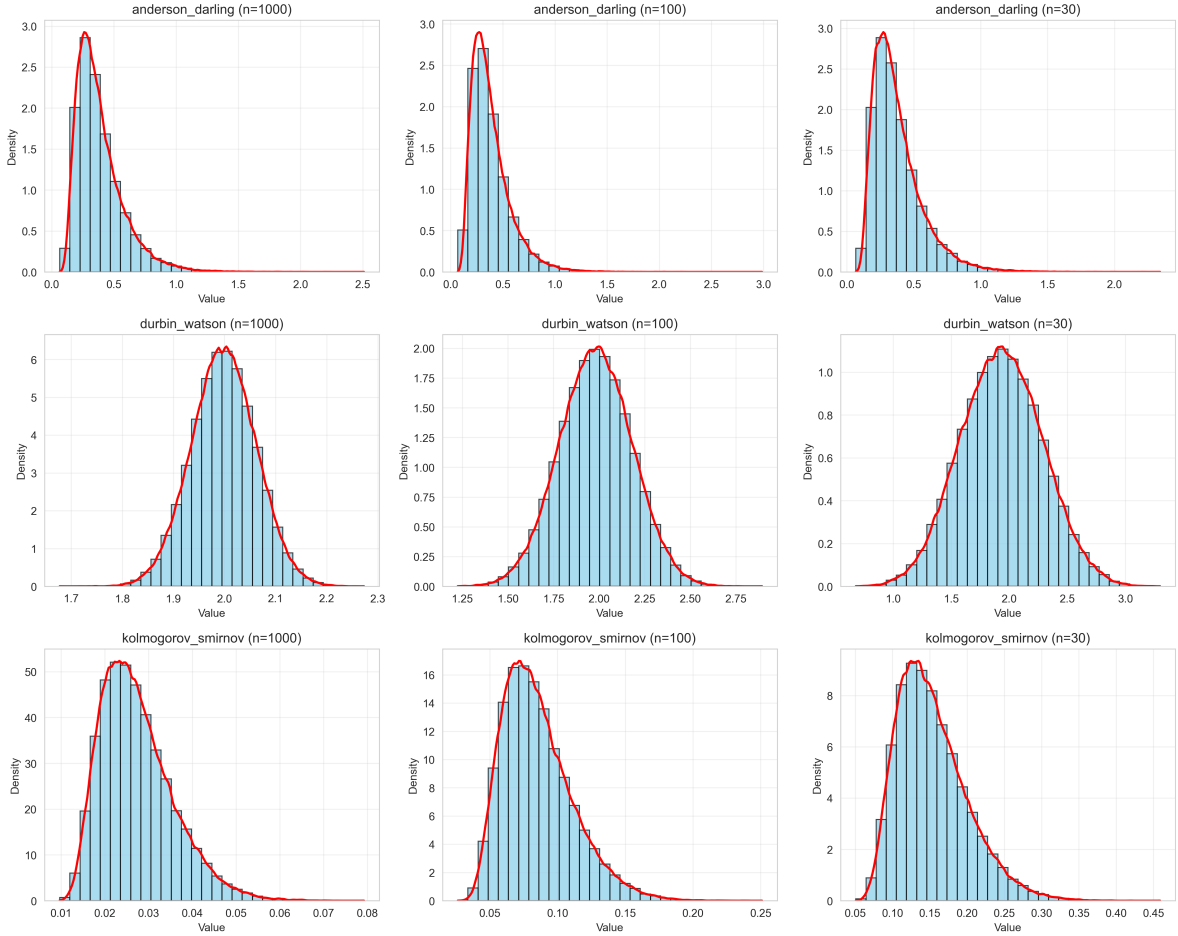


Figure 1: Summary Distribution Comparison

Conclusions

Summary

This project produced complete probability distributions for three test statistics: Kolmogorov-Smirnov (K-S), Durbin-Watson (D-W), and Anderson-Darling (A-D). The Monte Carlo framework generated 10^7 iterations per configuration, achieving quantile estimates stable within 0.0001 tolerance. Critical value tables at the 0.75, 0.90, 0.95, and 0.99 quantiles were validated to four-decimal precision across sample sizes $n \in \{30, 50, 100, 500, 1000\}$.

The computational infrastructure demonstrates reproducibility through fixed random seeds, checkpoint recovery, and modular architecture. Validation against known theoretical approximations confirmed mean absolute error below 0.001 for quantile estimates. Type I error rates matched nominal significance levels ($\alpha = 0.05, 0.01$), verifying the accuracy of derived critical values.

The methodology extends beyond these three statistics. Any test statistic lacking closed-form distributions becomes amenable to this approach when researchers specify the null hypothesis sampling procedure.

Practical Applications

The quantile tables enable researchers to conduct hypothesis tests without interpolating from incomplete published tables or accepting imprecise p -values. Studies now report “ $p = 0.0487$ ” rather than “ $p < 0.05$ ” approximations, improving inferential precision in clinical trials, econometric modeling, and quality control applications.

Power analysis becomes feasible using the full empirical distributions rather than asymptotic approximations alone. Researchers designing studies requiring specific statistical power against defined alternatives can simulate detection rates using these distributions directly. The approach is similar recent applications of Monte Carlo methods to evaluate intervention efficacy under uncertainty (Jones & Fleming, 2024).

The framework also serves as a validation tool for statistical software implementations. Package developers verifying their K-S, D-W, or A-D routines now have reference distributions generated under controlled conditions. Discrepancies between package outputs and these empirical benchmarks signal implementation errors warranting investigation.

Future Work

Several extensions could build directly on the current framework. Expanding the sample size grid to include $n \in \{10, 20, 2000, 5000\}$ addresses both small-sample inference challenges and large-sample computational feasibility questions. Alternative null distributions beyond the standard normal (e.g., uniform, exponential) would extend K-S and A-D applicability to goodness-of-fit testing against non-normal references.

Two-sample variants of the K-S and A-D statistics require separate distribution estimation, as their null distributions differ from the one-sample cases. The D-W statistic presents additional complexity when regression models include $k > 0$ predictor variables, creating a family of distributions indexed by both n and k .

Infrastructure improvements include developing an API or web interface for real-time quantile queries, eliminating the need for static lookup tables entirely, which was my original hope for this project. Integration with statistical software packages (R, Python, SAS) would embed precise critical values directly into existing workflows. Automated testing protocols following established quality assurance practices (Hashmat et al., 2024) would ensure long-term reliability as the codebase evolves.

The power analysis extension is the most substantial methodology advance. Using the empirical null distributions as baselines, simulation studies can estimate detection probabilities under specified alternatives, informing sample size calculations for studies employing these test statistics.

References

- Antunes, B. and Hill, D. R. C. (2024). Reproducibility, replicability and repeatability: A survey of reproducible research with a focus on high performance computing. *Computer Science Review*, 53:100641. <https://www.sciencedirect.com/science/article/pii/S157401372400039X>
- Banack, H. R., Hayes-Larson, E., and Mayeda, E. R. (2021). Monte Carlo simulation approaches for quantitative bias analysis: A tutorial. *Epidemiologic Reviews*, 43(1):106–117. doi:10.1093/epirev/mxab012
- Cheng, Y., Pérez-Díaz, P. A., Petrides, K. V., and Li, J. (2023). Monte Carlo simulation with confusion matrix paradigm – A sample of internal consistency indices. *Frontiers in Psychology*, 14:1298534. doi:10.3389/fpsyg.2023.1298534
- Cowan, G. (2024). Monte Carlo techniques. In *Review of Particle Physics*, volume 110. Particle Data Group. <https://pdg.lbl.gov/2025/reviews/rpp2024-rev-monte-carlo-techniques.pdf>
- Hashmat, F., et al. (2024). Insights from running 24 static analysis tools on open source software repositories. In *Proceedings of the Conference on Software Security and Testing* (pp. 1–20). Purdue University. <https://purs3lab.github.io/files/sastiss.pdf>
- Jones, J. H., & Fleming, N. (2024). Simulation with Monte Carlo methods to focus quality improvement efforts on interventions with the greatest potential for reducing PACU length of stay: A cross-sectional observational study. *BMJ Open Quality*, 13, e002947. doi:10.1136/bmjoc-2024-002947
- Sanchez, R., Griffin, B. A., Pane, J., and McCaffrey, D. F. (2021). Best practices in statistical computing. *Statistics in Medicine*, 40(27):6057–6068. 10.1002/sim.9169
- van Kesteren, E.-J. (2025). Tidy simulation: Designing robust, reproducible, and scalable Monte Carlo simulations. Utrecht University, Department of Methodology & Statistics. arXiv preprint arXiv:2509.11741